

## EXPERIMENT 40:

```
%{  
#include <stdio.h>  
#include <stdlib.h>  
%}  
  
%%  
[0-9]+[\+\-\*/][0-9]+ {  
    int a, b;  
    char op;  
  
    sscanf(yytext, "%d%c%d", &a, &op, &b);  
  
    switch(op)  
    {  
        case '+':  
            printf("Result = %d\n", a + b);  
            break;  
  
        case '-':  
            printf("Result = %d\n", a - b);  
            break;  
  
        case '*':  
            printf("Result = %d\n", a * b);  
            break;  
  
        case '/':  
            if(b == 0)  
                printf("Division by zero not allowed.\n");
```

```

        else

            printf("Result = %.2f\n", (float)a / b);

            break;

    }

}

```

```

\n    ;

```

```

.    ;

```

```

%%

```

```

int yywrap()
{
    return 1;
}

```

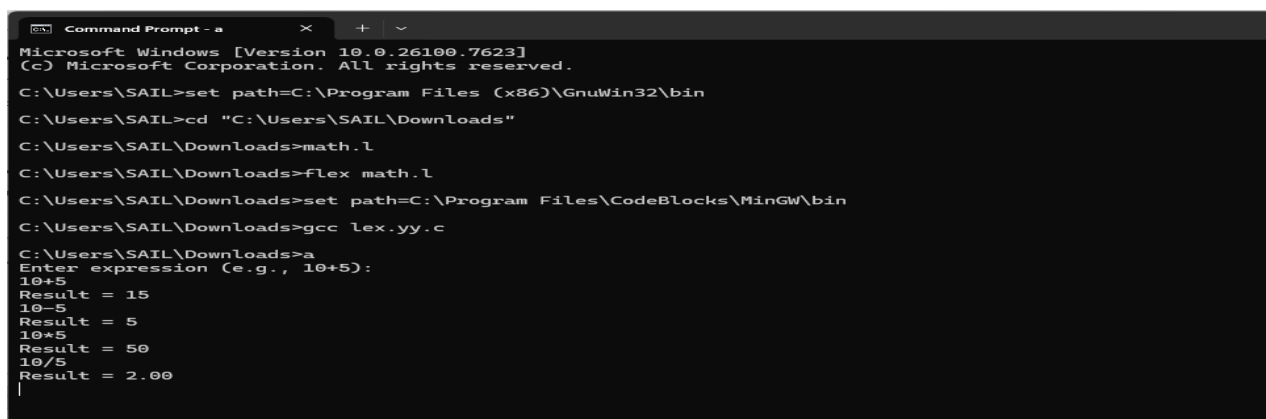
```

int main()
{
    printf("Enter expression (e.g., 10+5):\n");

    yylex();

    return 0;
}

```



```

Command Prompt - a
Microsoft Windows [Version 10.0.26100.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\SAIL>set path=C:\Program Files (x86)\GnuWin32\bin
C:\Users\SAIL>cd "C:\Users\SAIL\Downloads"
C:\Users\SAIL\Downloads>math.l
C:\Users\SAIL\Downloads>flex math.l
C:\Users\SAIL\Downloads>set path=C:\Program Files\CodeBlocks\MinGW\bin
C:\Users\SAIL\Downloads>gcc lex.yy.c
C:\Users\SAIL\Downloads>a
Enter expression (e.g., 10+5):
10+5
Result = 15
10-5
Result = 5
10*5
Result = 50
10/5
Result = 2.00
|

```